



Society of Petroleum Engineers

SPE-229138-MS

Innovative Low Co₂ Cement, Tailored for Performance in Deep Water Wells, Harsh Chemical Environment and CCS

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Microfine cement is used in the oilfield industry for their ability to seal low permeability rocks and fissures, as well as for their enhanced cementitious performance. In this paper, we present the use of Fine Cement blends with a high amount of inert material and environmentally friendly chemicals, allowing for a low CO₂ footprint, high chemical and mechanical resistance in low density slurries for deep water application and improved zonal isolation and alleviate logistics using a single blend system. The new cement blend is obviously usable for other cementing applications.

This study focuses on a system of a new microfine cement slurry incorporating micro silica, microfine cement, inert materials, and renewable and environmentally friendly chemicals. A typical 1.60 SG cement slurry was prepared using the API/ISO standard mixing protocol and tested using a viscometer, fluid loss, HPHT consistometer, ultrasonic cement analyzer, curing chamber and compression testing machine. Evaluations were conducted at low temperature (25 -30°C). The results from laboratory analyses are presented.

The new cement system demonstrates an overall lower CO₂ footprint, being three times lower than that of conventional Portland cement. The proposed system shows high early compressive strength (above 1000 psi) after 12 hours at sea bed temperature, and low permeability and set cement porosity. Furthermore, the scope of the tests aimed to develop a deepwater system that is eco-friendly and cost-effective, with the capability to significantly reduce operational costs. The combination of microfine cement, pozzolanic micro silica, inert materials, and environmentally friendly renewable chemicals results in a tight cement system that is well-suited for harsh chemical environments such as sour gas and CCS wells. Additionally, this new cement blend can be used in other renewable energy sectors such as geothermal, lithium extraction and hydrogen storage.

This paper presents a new cement system with lower CO₂e footprint compared to conventional Portland cement. The new cement designed with extremely low Portland cement content is suitable for primary and remedial cementing operations and in particular plug and abandonment. The system exhibits very low permeability, which makes it more durable compared to conventional cement for harsh chemical environments such as sour gas and CCS wells.

Introduction

Cementing operations are vital in well construction, full isolation of the annular space protects the casing from interacting with surrounding formations and the fluids they hold, which otherwise would lead to corrosion and integrity failure or well abandonment.

Today, the deepwater region is defined as a sea area with a depth greater than 500 meters. Cementing operations through low-temperature formations prone to shallow flows are becoming increasingly common in deepwater environments. Near-freezing temperatures, flowing sands, and lost circulation present significant challenges during critical surface casing running and cementing operations. The low temperatures can significantly slow the cement hydration process, compromising gel strength and compressive strength development—both of which are vital for preventing fluid migration and ensuring structural support for the well.

Industry collaboration among various operators and service companies has highlighted several key slurry and set-cement properties essential for mitigating shallow flow issues:

- Precise control of slurry density at both surface and downhole conditions
- Appropriate rheology to ensure effective mud displacement
- Rapid development of gel strength and compressive strength
- Minimal shrinkage and low permeability
- Tailored mechanical properties of set cement to ensure long-term hydraulic isolation

Lightweight cement slurries are critical in deepwater well construction due to the narrow pore-fracture pressure margins and the presence of weak, low-temperature formations. In these conditions, conventional-density cement slurries can exceed fracture gradients, leading to lost circulation, formation damage, or well control issues. Lightweight cement systems—typically ranging from 10.0 to 13.5 lb/gal (1.19 SG to 1.61 SG) help reduce hydrostatic pressure while still providing the mechanical integrity and zonal isolation necessary for long-term well stability.

Traditionally, extenders such as bentonite, pozzolans, or hollow glass microspheres are used to lower slurry density. However, these materials can present drawbacks, including poor long-term durability, increased permeability, and slower strength development in cold environments. This technical paper proposes the use of Microfine Cement as an alternative extender to address these limitations.

Microfine Cement, characterized by its ultra-fine particle size and high surface area, improves particle packing, reduces free fluid, and enhances early gel strength development. These properties contribute to a more stable slurry with better displacement efficiency and improved bonding. The denser, low-permeability set structure also minimizes shrinkage and fluid migration risks.

In addition to performance benefits, Microfine Cement offers logistical and environmental advantages. Its higher efficiency reduces the total volume of materials required at the wellsite, leading to lower transportation and storage costs—critical in remote deepwater operations. Furthermore, the reduced use of chemical extenders and optimized material usage contribute to a lower environmental footprint, aligning with industry goals for more sustainable operations.

Cement and Chemicals Carbon Footprint

The analysis of CO₂ emissions can be complex, ranging from examining the initial effects to tracing back to the extraction of raw materials. However, for practicality and clarity, this analysis focuses solely on the primary level of impact. This includes the reduction in CO₂ emissions achieved through savings on the use of chemical products. It does not extend to the analysis of the supply chain responsible for providing these

chemical products, logistics, and other components, as we considered equal to the conventional chemicals currently used.

The cement industry is one of the largest producers of CO₂ in the world. According to IEA data [IEA(2023)], in 2022, the cement industry was responsible for 2.5Gt of CO₂ emissions which represents a share of 27% of the industrial emissions of CO₂ (7.6Gt) and a share of 7% of the total manmade CO₂ emissions (29Gt). The cement manufacturing process requires large amounts of thermal and electrical energy. This energy requirement combined with the widespread use of concrete in global construction defines the magnitude of the challenge at hand. The estimated CO₂ production for the Portland cement is 0.85-1.00 ton of CO₂ per ton of cement produced. The most commonly emission are coming from the Calcination process that represent the 50-60%, the fuel combustion 30-40% due to the high temperature that the clinker formation and 5-10% for the electricity.

The second largest CO₂ producer is the chemicals industry right after the Cement with a 25% share if industrial emissions with a total of 2.3Gt of CO₂ emission.

Cement and chemical industries are among the largest contributors to global CO₂ emissions, making their decarbonization crucial for climate goals. Cement production releases CO₂ primarily through limestone calcination and fuel combustion, requiring innovative solutions like clinker substitution, carbon capture, and alternative binders. Similarly, the chemical sector depends heavily on fossil fuels for both energy and feedstock, with ammonia and ethylene production being major emission sources. Transitioning to electrification, green hydrogen, and sustainable feedstocks can significantly lower their carbon footprint. Rapid CO₂ reduction in these sectors is not just an environmental necessity but also a step toward achieving net-zero targets and ensuring a sustainable future.

Achieving a Low Product Carbon Footprint System

Composition and Technical Features

Low Carbon Footprint Superfine Cement System (LCFSCS) is an innovative material that blends finely ground cementitious components to create a high-performance, low-thickness surface coating. The incorporation of OPC cement and blast furnace slag, prior to a precise milling, in Superfine Cement System formulation presents significant structural and environmental advantages. LCFSCS made with Class G cement and blast furnace waste consists of Class G cement, which is commonly used in oil well applications and is characterized by its high sulfate resistance and low water-to-cement ratio, providing enhanced durability. Blast Furnace Waste, specifically Granulated Blast Furnace Slag (GBFS), is a byproduct of steel production and serves as a pozzolanic material that improves the microstructure of cement by reacting with calcium hydroxide to form additional calcium silicate hydrates (C-S-H), the primary strength-bearing component in cementitious materials. Addition of slag has also proven to be effective against corrosive environments and even carbonation, which is key in CCUS wells. Additionally, micron size fine and large amount of inert crystalline silica sand and conventional oilfield cementing additives ensure smooth application, enhanced workability, and improved bonding. The microstructure of this composite material is dense, with refined pore distribution. The extra pozzolanic reaction between GBFS and Class G cement results in improved hydration kinetics, leading to greater compressive strength and reduced permeability.

To further enhance its performance, Superfine Cement System formulations can incorporate superfine, high-reactivity cement with high slag content, combined with a high amount of pozzolanic crystalline silica (up to 100 %BWOC). The addition of microfine pozzolanic Microsilica or Silica Fumes particles significantly improves slurry and set-cement performance by offering gas-migration control, reducing set-cement porosity, and decreasing permeability. These modifications result in a denser, more durable material with superior mechanical properties, making it ideal for high-performance applications.

Another advantage of this formulation is its superior performance in aggressive environments. Due to the reduced permeability and refined pore structure, Superfine Cement System with GBFS demonstrates increased resistance to chemical attacks, sulfate exposure, and chloride penetration. This makes it ideal for marine structures, oil and gas applications, and high-moisture environments where conventional cement materials may deteriorate more rapidly. The improved durability further contributes to sustainability by extending the lifespan of structures and minimizing the need for frequent maintenance, repairs, or replacements.

Environmental and Sustainability Benefits

Beyond structural advantages, integrating GBFS (Ground Granulated Blast Furnace Slag) into the Superfine Cement System delivers significant sustainability gains. Cement production is a major CO₂ emitter, largely from energy-intensive clinker manufacturing. By replacing part of the clinker with GBFS—a steel industry byproduct—the process:

- Cuts CO₂ emissions significantly.
- Repurposes industrial waste, supporting a circular economy.
- Reduces the need for virgin raw materials.

This approach also enhances mechanical strength and resistance to harsh environments while remaining compatible with standard cement production for large-scale use. In oilfield applications, GBFS blends:

- Achieve up to 70% lower PCF (per barrel) than neat Portland cements.
- Match other technologies in low CO₂ output (200–300 kg CO₂/MT, EXW basis).
- Avoid corrosive activators and keep full compatibility with conventional cementing chemicals and equipment.

By adopting this formulation, industries lower their carbon footprint, manage waste efficiently, and produce durable, eco-friendly construction materials—marking a practical step toward modern sustainable construction.

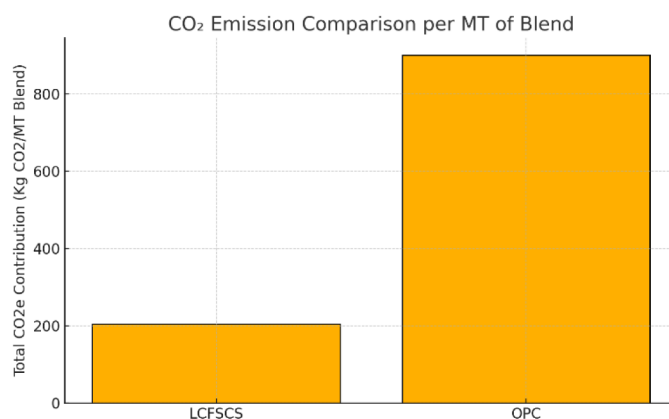


Figure 1—CO₂ Emission Comparison per MT of Blend

Comparative Performance and Carbon Footprint Analysis

In this paper, we present a comparative evaluation of various 1.60 SG (13.4 PPG) cement slurry systems designed for deepwater applications. The study includes a novel formulation based on a superfine, high-reactivity cement with elevated slag content, combined with up to 100% BWOC of pozzolanic crystalline

silica and enhanced with microfine pozzolanic additives such as Microsilica or Silica Fume. This engineered blend delivers slurry and set-cement performance that is comparable to or better than conventional Lightweight Agent (LWA) cement slurries, particularly in terms of gas migration control, reduced porosity, and low permeability. Laboratory testing confirmed its ability to meet deepwater operational requirements, with consistent early strength development, slurry stability, and zonal isolation. Additionally, the system features a low CO₂e footprint, is easy to prepare on-site using just three main components, remains compatible with standard oilfield cementing additives, and is particularly suitable for low-volume operations such as plug and abandonment (P&A). The blend's minimal Portland cement content contributes to a low heat of hydration, making it well-suited for thermally sensitive deepwater formations.

The superfine cement at the base of this slurry system is based on a Portland clinker, has a PCF of 389 kg CO₂/MT EXW thanks to its inherent high slag content—a by-product from the steel industry with virtually 0 PCF ex-plant. During the superfine cement manufacturing, slag is post-added to the Portland cement fraction before being milled precisely to size, eliminating any CO₂ from competing calcination processes e.g., API Class L Cement. Crystalline Silica (sand and flour) sourced locally in bulk have typical PCFs of 30 kg CO₂/MT (check with ECS silica sand supplier). The sub-micron pozzolanic material known as Microsilica or silica fumes is available as powder or liquid suspension and has a very low PCF of 36 kg CO₂/MT, being a by-product of the metal industry.

The PCF of the cement slurry design presented here is only 204 MT CO₂e per 1 MT of dry blend. This is ~78% less than a neat Portland cement equivalent system at the same density. Reference Portland G cement is often reported at 900 kg CO₂ per MT in Europe and more than 1 MT in USA. Since the presented cement blend system exhibits a slightly lower water content (53% slurry porosity) compared to its equivalent similar density OPC system (58%), the PCF of this new cement blend system is therefore 70% lower than its OPC equivalent on a per BBL of slurry pumped.

Table 1—Table the different CO₂ emission for each system proposed in the paper with a comparison for MT of Blend.

Table 1—CO₂ Emission Comparison for MT of Blend

Blend Composition	Blend Composition %	PCF in CO ₂ /MT *EXW for Material	CO ₂ e Contribution In Kg CO ₂ /MT of Blend
LCFSCS:			
Super Fine Cement	48.5 %	389	188.6
Silica Sand	48.5 %	30	14.5
Microsilica dry	3.0 %	36	1.0
TOTAL:	100%	455	204
OPC:			
Portland Class G	100 %	900	900

*EXW PCF are considered for comparison purpose vs. OPC systems. Logistics related CO₂ emissions must be added for a true evaluation of final PCF at the location.

Alternative Chemicals

The shift from conventional petrochemicals to bio-based and renewable alternatives is essential for reducing carbon emissions and dependence on fossil fuels. Conventional petrochemicals, derived from crude oil and natural gas, are widely used in plastics, fertilizers, and industrial chemicals but contribute significantly to greenhouse gas emissions. Transitioning to bio-based and renewable sources involves adopting sustainable feedstocks, innovative production methods, and alternative energy sources.

One key approach is replacing fossil-based feedstocks with bio-based materials such as plant-derived sugars, lignocellulosic biomass, and algae. These renewable resources can be processed into essential

chemicals like bio-ethylene, bio-methanol, and bio-ammonia, reducing carbon intensity while maintaining similar functionalities. Additionally, fermentation and biocatalysis enable the production of bioplastics and bio-based polymers, which offer biodegradable and sustainable alternatives to petroleum-based plastics.

Chemicals used for this project are mainly foam preventers, fluid loss, cement dispersants and retarders. Figure 2 presents a summary of the eco-friendly chemical additives selected for the project. Each additive is described in terms of its function, formulation characteristics, and environmental benefits. The chosen materials emphasize low-VOC content, biodegradability, and the use of renewable resources, aligning with sustainable manufacturing practices and green certification requirements.

Chemicals / Function	Description	Eco-Friendly Features
Foam Preventer	Silicone-free, liquid defoamer designed for water-based systems, a strong focus on eco-friendliness. It enhances foam control without contributing to volatile organic compound (VOC) emissions suitable for low-VOC and environmentally-friendly formulations.	Supports green manufacturing practices, excellent industrial system compatibility, and long-lasting foam suppression
Dispersant	VOC-free dispersing and wetting agent designed for environmentally conscious formulations, enhances dispersion, reduces viscosity, and improves strength.	No APEOs or VOCs or VOCs meets green certification standards, optimizes aqueous system compatibility
Fluid Loss (Hydroxyethyl Cellulose)	Hydroxyethyl Cellulose (HEC) is a naturally derived, biodegradable polymer obtained from plant-based cellulose.	Renewable origin and biodegradable Free from synthetic additives Low-impact formulation
Retarder	An advanced, eco-friendly additive derived from renewable lignin, for use in water-based fluids as a sustainable alternative to synthetic polymers, improves rheology	Lignin-based formulation Minimal reliance on petroleum-based products

Figure 2—A summary of the eco-friendly chemical additives selected for the project

Laboratory Studies

The American Petroleum Institute (API) defines key performance requirements for cement slurries under downhole conditions and provides standardized testing methods. Critical parameters include density, rheology, static gel strength, thickening time, fluid loss, compressive strength, and free water.

Slurry Preparation

The target slurry density was 1.90 SG (15.8 ppg). Prepared in an API blender, it was mixed for 15 s at 4,000 rpm and 35 s at 12,000 rpm (Nelson, 1990).

Rheology Evaluation

Conditioned in an atmospheric consistometer, the slurry's viscosity was measured with a Fann Model-35 viscometer. Plastic viscosity and yield point determined flow behavior, suspension capability, and prevention of solids settling or fluid loss. Proper rheology is essential for effective placement and stability.

Thickening Time

Measured with an API HP/HT consistometer, this test determines pumpable time before setting under simulated downhole temperature and pressure. Controlled setting time ensures proper placement, zonal isolation, and wellbore integrity.

Free Water

Using a 250 ml graduated cylinder over two hours, free water testing quantified water separation. Excess water can reduce cement integrity, increase permeability, and impair bonding. Density comparison across cured cement sections assessed slurry homogeneity.

Compressive Strength

Evaluated with an Ultrasonic Cement Analyzer (UCA), this test continuously measured strength development. Adequate compressive strength ensures resistance to downhole stresses, durability of the cement sheath, and prevention of cracking or debonding. Optimized formulations achieve the required strength for wellbore applications.

Case Study

As the oil and gas industry strives to reduce its carbon footprint, the transition toward eco-friendly deepwater cementing solutions has gained momentum. This case study presents a comparative evaluation of a conventional slurry versus a more sustainable alternative.

The conventional slurries were prepared using Portland Cement and petrochemical-based additives, while the alternative slurry was formulated with Low CO₂ Cement (LCC) and bio-sourced additives, reflecting the industry's shift toward greener technologies.

The primary objective of this study was to assess the performance, environmental impact, and operational feasibility of six formulations under real well conditions. The well, characterized by a bottom hole static temperature (BHST) of 30°C and a bottom hole circulating temperature (BHCT) of 12°C, provided an ideal setting to evaluate the thermal stability, mechanical integrity, rheology, and fluid loss control of all slurries. The bio-sourced formulation was selected to reduce CO₂ emissions and reliance on fossil-derived additives, aligning with global carbon reduction initiatives.

The results presented in this case study offer insights into the potential of low-CO₂, bio-based cement slurries to maintain or exceed the performance of conventional systems, while contributing to the broader sustainability goals of the energy sector.

The well type chosen for this study is situated in the West Africa oceans, with a sea depth of 1250 m.

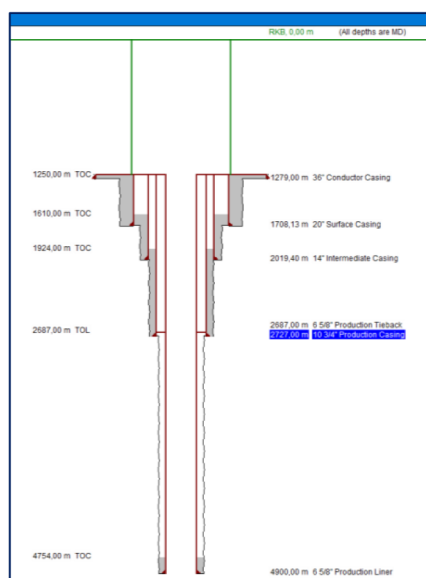


Figure 3—Well Schematics.

The new approach cement system was designed and tested according with the data in Table 1. The objective was to replicate a cement slurry system with low CO₂ impact and eco-friendly that will respect the same objective of cement placement in conventional conditions.

Experimental Conditions

All the slurries were mixed and tested using current API standards and procedure.

Table 2—Table shows the testing condition for the different slurries' formulation. The only variable during all the study was the slurry composition, to compare the slurries characteristics.

Table 2—LABORATORY TESTING CONDITION

	BHST (°C)	BHCT (°C)	Pressure (psi)	Slurry Density (SG)	Mix Water Fluid
CONVENTIONAL SLURRIES	30	12	1000	1,60	Fresh Water
ECO-FRIENDLY SLURRIES	30	12	1000	1,60	Fresh Water & Sea Water

Results

Table 4 shows the thickening time of each test, fluid loss, compressive strength, and rheology of the cement systems. The first parameter that we should consider is the test thickening time which has been kept constant and targeting a range of 5 to 5,5 hours without.

Compressive strength was also taken into consideration as an important parameter during this study to investigate a possible improvement in the early development of compressive strength to anticipate and cut the wait on cement time (WOC) during cementing operation. In deepwater operations, the Wait on Cement (WOC) period is a critical phase that directly impacts both well integrity and project economics. Low seabed temperatures slow cement hydration, requiring longer WOC times to reach sufficient compressive strength. Delays in this phase can significantly increase rig time and associated costs, with deepwater rig rates reaching hundreds of thousands of dollars per day. However, resuming operations too early risks poor zonal isolation, gas migration, or casing failure. Optimizing WOC is therefore essential—not only to ensure cement performance and long-term well stability but also to reduce non-productive time and control operational costs.

Cement slurry rheology tests are crucial for evaluating flow behavior, stability, and pumpability, ensuring efficient placement in construction and oil well applications. Comparing the six different formulations helps optimize viscosity, yield stress, and settling resistance, reducing operational risks and enhancing performance in varying environmental conditions. Proper testing ensures durability and structural integrity.

SLURRY DESIGN TEST RESULTS	Slurry 1, 60SG NEAT	Slurry 1, 60SG with Bentonite	Slurry 1, 60SG with Sodium Silicate (AS-3L)	Slurry 1, 60SG Lightweight (with ULW-6000) and Fresh water	Slurry 1, 60SG with (MCC-3.0) and Fresh water	Slurry 1, 60SG with (MCC-3.0) and Sea water	Slurry 1, 60SG with (70% MCC-3.0+30% G) and Sea water	Slurry 1, 60SG with (MCC-3.0) and Sea water
Rheology Test	Vp=14 (Cp) Ty=17 (lb/100 ft ²)	Vp=9 (Cp) Ty=26 (lb/100 ft ²)	Vp=30 (Cp) Ty=10 (lb/100 ft ²)	Vp=165 (Cp) Ty=35 (lb/100 ft ²)	Vp=121 (Cp) Ty=29 (lb/100 ft ²)	Vp=33 (Cp) Ty=16 (lb/100 ft ²)	Vp=20 (Cp) Ty=6 (lb/100 ft ²)	Vp=30 (Cp) Ty=15 (lb/100 ft ²)
Free water	7 ml	4 ml	4 ml	0 ml	0 ml	0 ml	0 ml	0 ml
Fluid Loss (API)	N/A	N/A	N/A	55 ml	49 ml	58 ml	62 ml	80 ml
Thickening Time	30B=4h29 60B=6h15 100B=6h02	30B=5h44 60B=7h05 100B=6h29	30B=4h34 60B=4h47 100B=4h52	30B=N/A 60B=4h24 100B=4h51	30B=2h39 60B=4h47 100B=6h09	30B=2h30 60B=3h05 100B=4h23	30B=5h37 60B=6h48 100B=7h56	30B=4h48 60B=5h25 100B=6h43
UCV/SCSA	50psi=07h00 500psi=N/A 24h=342 psi	50psi=6h33 500psi=N/A 24h=346 psi	50psi=6h57 500psi=20h14 24h=619 psi	50psi=6h07 500psi=11h56 24h=1236 psi	50psi=7h57 500psi=13h34 24h=1140 psi	50psi=11h16 500psi=19h51 24h=967psi 48h=3114psi 72h=3700psi	50psi=14h29 500psi=23h59 24h=504psi 48h=3342psi 72h=4059psi	50psi=13h38 500psi=21h22 24h=861psi 48h=2613psi 72h=3403psi

Figure 4—Table shows the laboratory test results performed with the six different slurries.

The chart presents a comparative analysis of various 1.60 SG (13.4 PPG) cement slurry systems used in deepwater conditions, focusing on key performance metrics such as rheology, fluid loss, thickening time, and compressive strength development. The evaluation highlights both conventional and advanced slurry designs incorporating different additives and mix waters.

Conventional systems, such as neat cement and formulations with bentonite or sodium silicate (AS-3L), exhibited high fluid loss, extended thickening times, and relatively slow strength development. For example, the sodium silicate-based slurry reached only 619 psi at 24 hours, indicating limited early compressive strength, a critical factor for wait-on-cement (WOC) decisions in deepwater operations.

In contrast, slurries designed with Microfine Cement Components (MCC-3.0)—particularly those combined with seawater and pozzolanic materials—showed significantly improved performance. The MCC-3.0-based slurry with freshwater demonstrated excellent rheology ($V_p = 121$ cP, $T_y = 29$ lb/100 ft²), zero free water, low fluid loss (49 ml), and a rapid strength buildup (1140 psi at 24h). When combined with seawater, MCC-3.0 slurries maintained stability and continued to outperform conventional systems. Notably, the 70% MCC-3.0 + 30% G blend achieved over 4000 psi at 72 hours, with 500 psi reached in just under 24 hours.

These results suggest that replacing traditional lightweight agents and extenders with Microfine Cement can improve set-cement properties, reduce WOC time, and enhance operational efficiency. Furthermore, MCC-3.0 systems offer robust performance even in saline environments, supporting offshore and deepwater applications where seawater is the only viable mixing fluid. The consistent and accelerated strength development across all MCC-based systems indicates a strong potential for optimizing deepwater cementing designs while minimizing environmental impact and logistical complexity.

Conclusion

- The incorporation of Class G cement and Granulated Blast Furnace Slag (GBFS) in Superfine Cement System formulations significantly enhances durability, compressive strength, and resistance to aggressive environmental conditions.
- The reaction between GBFS and Class G cement leads to a denser microstructure, reduced permeability, and refined pore distribution, contributing to superior mechanical properties and extended service life.
- The Compressive strength of the new cement system showed a greater value compared with the conventional Portland Class G Cement.
- Utilizing in Superfine Cement System reduces the environmental impact of cement production by lowering CO₂ emissions, conserving raw materials, and promoting sustainable construction practices.
- The use of industrial byproducts, supports circular economy principles by repurposing steel industry waste into valuable construction materials, reducing landfill waste, and promoting resource efficiency.
- Microfine Cement (MCC-3.0) slurries demonstrated faster early strength development compared to conventional slurries.
- MCC-based systems performed effectively even when mixed with seawater, ensuring suitability for offshore and deepwater use, and slurries showed minimal free water and controlled fluid loss, enhancing zonal isolation and reducing gas migration risk.
- Faster compressive strength gain leads to shorter Wait-On-Cement (WOC) times and lower rig-related costs.
- MCC blends, especially with partial Class G cement, achieved high long-term compressive strengths (>4000 psi at 72h).

- Widespread adoption of Superfine Cement System with GBFS can transform the cement industry by offering a durable, low-carbon alternative that aligns with global sustainability goals and advances green building initiatives.

Nomenclature

API	American Petroleum Institute
ISO	International Organization for Standardization
HPHT	High Pressure High Temperature
UCA	Ultrasonic Cement Analyzer
CO₂	Carbon Dioxide
CCS	Carbon Capture and Storage
CO_{2e}	Carbon Dioxide Equivalent
OPC	Ordinary Portland Cement
LCFSCS	Low Carbon Footprint Superfine Cement System
GBFS	Granulated Blast Furnace Slag
C-S-H	Calcium Silicate Hydrate
PCF	Product Carbon Footprint
BWOC	By Weight of Cement
LWA	Lightweight Agent
P&A	Plug and Abandonment
MT	Metric Ton
EXW	Ex Works (Incoterm)
LCC	Low CO ₂ Cement
BHST	Bottom Hole Static Temperature
BHCT	Bottom Hole Circulating Temperature
WOC	Wait on Cement
VOC	Volatile Organic Compound

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